HARI K S

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Salem, Tamil Nadu, India

SUMMARY

Electronics and Communication Engineering undergraduate focused on Automotive Electronics and Embedded Systems. Experienced in microcontroller programming, sensor-actuator interfacing, and serial communication protocols (UART, SPI, I2C). Seeking an entry-level engineering role.

EDUCATION

Knowledge Institute of Technology, Salem, Tamil Nadu 2023 – 2027
B.E. in Electronics and Communication Engineering | CGPA: 7.34 / 10.0

Notre Dame of Holy Cross School, Salem, Tamil Nadu 2022
Class 12 (Higher Secondary) | 63.6%

Notre Dame of Holy Cross School, Salem, Tamil Nadu 2020
Class 10 (Secondary) | 83.4%

TECHNICAL SKILLS

Languages: C, Embedded C, Java
Microcontrollers: ESP32, ESP8266, Arduino UNO, STM32
Protocols: UART, SPI, I2C, GPIO
Tools & IDEs: STM32CubeIDE, Arduino IDE, VS Code, GitHub

INTERNSHIPS

Sans Innovations, Salem Feb 2024
Embedded System Design Intern
- Wrote firmware to interface peripheral sensors using UART, I2C, and SPI protocols.
- Tested and verified program execution and signal stability on development boards.

Codec Technologies Mar 2026 – Apr 2026
Digital Electronics & VLSI Intern
- Worked on digital logic design, combinational circuits, and sequential circuit analysis.
- Studied fundamental digital verification workflows and IC design methodologies.

PROJECTS

Rail Bot – Track Inspection Prototype
Tech Stack: Arduino, Ultrasonic Sensor, Motor Driver, DC Motors
- Constructed a motorized track vehicle to inspect rails and identify physical obstacles.
- Implemented continuous distance measurement to automatically trigger motor braking upon detecting an obstruction.

EcoCare – Child Safety & Location Tracker Aug 2025
Tech Stack: ESP32, GSM (SIM800L), GPS (NEO-6M), Embedded C
- Developed a wearable prototype to acquire GPS coordinates and transmit location alerts via SMS.
- Configured UART communication to parse standard NMEA location strings reliably under weak signal conditions.

Smart RFID Door Lock Aug 2024
Tech Stack: ESP8266, RC522 RFID Module, Relay Module
- Built an access control system using RFID cards communicating over SPI.
- Programmed microcontroller logic to validate card UIDs and switch a 5V relay for door actuation.

CERTIFICATIONS

- **Microsensors and Nanosensors** – NPTEL Certification (Elite + Silver)
- **Sensors and Actuators** – NPTEL Certification (Elite)
- **Cloud Computing** – NPTEL Certification (Elite)

ACHIEVEMENTS

- **Finalist** – Shortlisted for Grand Finale in Techzooka Hackathon organized by IEEE (2024).

LANGUAGES

Tamil: Native (Speaking, Reading, Writing)
English: Professional Working Proficiency (Speaking, Reading, Writing)